MINISTRY OF STEEL • GOVT. OF INDIA

MOIL LIMITED

NATIONAL MANGANESE MINING INTELLIGENCE PLATFORM

ANNUAL ENTERPRISE OPERATIONAL & AI MULTI-MINE ASSESSMENT

Comprehensive Telemetry, SCADA Machinery, Geostatistical Reserve & Stress Simulation Report

Report Generated:	27 August 2026, 10:07:02 IST
Intelligence Engine:	FastAPI Gateway + 6 Production ML Models (Shortfall-GBM, Reserve-RF, Equip-GBM, Anomaly-IF)
Coverage Scope:	10 MOIL Assets (3 Madhya Pradesh, 7 Maharashtra)
Statutory Oversight:	DGMS Metalliferous Mines Regulations (MMR 1961) Compliant
Classification:	Synthetic Demonstration & Operational Digital Twin

1. Executive Summary & National Portfolio Overview

MOIL Limited operates the premier underground and opencast manganese extraction assets across the Central Indian Sausar Belt. This report synthesizes real-time SCADA telemetry, geostatistical block prospectivity, equipment health degradation curves, and 40 stress simulation scenarios across all 10 mining centers.

TOTAL MINES	ALLOCATED TARGET	24H YIELD FORECAST	STRESS SHORTFALL	AVG Mn GRADE	AI TRUST SCORE
10 Sites 3 MP, 7 MH	**32,400 T** Per Day	**25,322 T** (78.2% Yield)	**-7,078 T** (-21.8% Risk)	**41.8% Mn** High Metallurgical	**95.8%** Bayesian Trust

2. Canonical 10-Mine Operational Matrix

Mine Asset	State / District	Type	Coordinates (WGS84)	Target (TPD)	Grade (% Mn)	Fleet	AI Trust
Balaghat Mine	Madhya Pradesh (Balaghat)	Underground Deep Shaft	21.8499°N 80.2267°E	6,200 T	44.2%	32 Units	95.4%
Tirodi Mine	Madhya Pradesh (Balaghat)	Opencast & Incline Shaft	21.6889°N 79.7028°E	3,100 T	39.4%	24 Units	95.4%
Ukwa Mine	Madhya Pradesh (Balaghat)	Underground Drift & Adit	21.9747°N 80.4600°E	1,850 T	41.8%	18 Units	95.4%
Munsar Mine	Maharashtra (Nagpur)	Underground Deep Shaft	21.3972°N 79.2889°E	2,400 T	38.6%	20 Units	95.4%
Kandri Mine	Maharashtra (Nagpur)	Semi-Mechanized Underground Shaft	21.4194°N 79.2667°E	2,800 T	42.5%	22 Units	95.4%
Gumgaon Mine	Maharashtra (Nagpur)	Deep Vertical Shaft	21.3917°N 79.0167°E	3,400 T	43.8%	26 Units	95.4%
Chikla Mine	Maharashtra (Bhandara)	Underground Sub-Level Stoping	21.5500°N 79.7667°E	4,100 T	41.2%	28 Units	95.4%
Dongri Buzurg Mine	Maharashtra (Bhandara)	Large Opencast Bench & EMD Plant	21.5583°N 79.6833°E	5,400 T	40.5%	36 Units	95.4%
Ramtek Operations	Maharashtra (Nagpur)	Mechanized Opencast & Infill	21.4000°N 79.3333°E	1,600 T	37.8%	16 Units	95.4%
Bhandara Mine (Beldongri)	Maharashtra (Bhandara)	Underground & Opencast Cluster	21.4500°N 79.5167°E	1,950 T	38.2%	18 Units	95.4%

3. Detailed Mine Asset Profiles & Telemetry Baselines

Balaghat Mine (Madhya Pradesh) — Underground Deep Shaft

District & State:	Balaghat, Madhya Pradesh	**Daily Committed Target:**	6,200 Tonnes/Day
WGS84 Coordinates:	21°50'59.8"N 80°13'36.2"E	**Ore Grade & Quality:**	44.2% Mn (High Grade)
Elevation / Water Table:	330m MSL (-185m Level)	**SCADA Sensor Network:**	184 Telemetry Channels
Heavy Fleet Asset Count:	32 Active Heavy Mobile Units	**Rainfall Sensitivity Factor:**	1.35x Inundation Multiplier

Tirodi Mine (Madhya Pradesh) — Opencast & Incline Shaft

District & State:	Balaghat, Madhya Pradesh	**Daily Committed Target:**	3,100 Tonnes/Day
WGS84 Coordinates:	21°41'20.0"N 79°42'10.0"E	**Ore Grade & Quality:**	39.4% Mn (Medium Grade)
Elevation / Water Table:	312m MSL (-95m Level)	**SCADA Sensor Network:**	112 Telemetry Channels
Heavy Fleet Asset Count:	24 Active Heavy Mobile Units	**Rainfall Sensitivity Factor:**	1.48x Inundation Multiplier

Ukwa Mine (Madhya Pradesh) — Underground Drift & Adit

District & State:	Balaghat, Madhya Pradesh	**Daily Committed Target:**	1,850 Tonnes/Day
WGS84 Coordinates:	21°58'28.8"N 80°27'36.0"E	**Ore Grade & Quality:**	41.8% Mn (Low Phosphorus)
Elevation / Water Table:	460m MSL (-120m Level)	**SCADA Sensor Network:**	96 Telemetry Channels
Heavy Fleet Asset Count:	18 Active Heavy Mobile Units	**Rainfall Sensitivity Factor:**	1.2x Inundation Multiplier

Munsar Mine (Maharashtra) — Underground Deep Shaft

District & State:	Nagpur, Maharashtra	**Daily Committed Target:**	2,400 Tonnes/Day
WGS84 Coordinates:	21°23'50.0"N 79°17'20.0"E	**Ore Grade & Quality:**	38.6% Mn (Standard Ferromanganese)
Elevation / Water Table:	310m MSL (-140m Level)	**SCADA Sensor Network:**	108 Telemetry Channels
Heavy Fleet Asset Count:	20 Active Heavy Mobile Units	**Rainfall Sensitivity Factor:**	1.3x Inundation Multiplier

Kandri Mine (Maharashtra) — Semi-Mechanized Underground Shaft

District & State:	Nagpur, Maharashtra	**Daily Committed Target:**	2,800 Tonnes/Day
WGS84 Coordinates:	21°25'10.0"N 79°16'00.0"E	**Ore Grade & Quality:**	42.5% Mn (High Grade Lumpy)
Elevation / Water Table:	318m MSL (-160m Level)	**SCADA Sensor Network:**	118 Telemetry Channels
Heavy Fleet Asset Count:	22 Active Heavy Mobile Units	**Rainfall Sensitivity Factor:**	1.28x Inundation Multiplier

4. 40-Permutation Scenario Stress & Recovery Matrix

Each mine possesses unique hydrogeological and structural characteristics. The table below details the deterministic simulated yield losses and shortfall mitigation actions calculated under 4 operational stress scenarios.

Mine Center	Baseline Yield	Monsoon Loss	Crusher Loss	Multi-Risk Loss	Max Loss %	Prescription Protocol
Balaghat	6,200 T	-1,382 T	-1,116 T	**-2,387 T**	**38.5%**	PROTO-BAL-01
Tirodi	3,100 T	-756 T	-558 T	**-1,308 T**	**42.2%**	PROTO-TIR-01
Ukwa	1,850 T	-366 T	-333 T	**-632 T**	**34.2%**	PROTO-UKW-01
Munsar	2,400 T	-513 T	-432 T	**-890 T**	**37.1%**	PROTO-MUN-01
Kandri	2,800 T	-590 T	-504 T	**-1,022 T**	**36.5%**	PROTO-KAN-01
Gumgaon	3,400 T	-795 T	-612 T	**-1,377 T**	**40.5%**	PROTO-GUM-01
Chikla	4,100 T	-906 T	-738 T	**-1,566 T**	**38.2%**	PROTO-CHI-01
Dongri Buzurg	5,400 T	-1,382 T	-972 T	**-2,386 T**	**44.2%**	PROTO-DON-01
Ramtek	1,600 T	-364 T	-288 T	**-628 T**	**39.2%**	PROTO-RAM-01
Bhandara	1,950 T	-425 T	-351 T	**-733 T**	**37.6%**	PROTO-BHA-01

5. AI Governance, DGMS Safety & Statutory Compliance

Human-in-the-Loop Statutory Mandate: All recommendations generated by the TreeSHAP explainability engine and Pareto optimizer serve strictly as operational decision-support advisories. Authorization of automated dispatch or mechanical throttling requires explicit sign-off by the certified Shift Mining Engineer under DGMS Metalliferous Mines Regulations.

Model Provenance & Calibration: Shortfall-GBM v1.0, Reserve-RF v1.0, Equipment-GBM v1.0, and IsolationForest Anomaly Detector maintain continuous Bayesian calibration across 284 SCADA channels with an aggregate confidence of 94.8% and sensor completeness of 98.2%.